

Ex

Consider the 1-D ODE

$$\dot{x} = f(x),$$

where $x(t) \in \mathbb{R}$, and $f: \mathbb{R} \rightarrow \mathbb{R}$ is infinitely differentiable (C^∞).

Suppose that $x_e \in \mathbb{R}$ is an equilibrium, i.e., $f(x_e) = 0$.

Some cases

1, $f'(x_e) > 0$.

$$f(x) - \underbrace{f(x_e)}_0 = f'(x_e) \cdot \underbrace{(x - x_e)}_{\Delta x} + \underbrace{R(\Delta x)}_{\text{higher-order terms}},$$

where $\frac{R(\Delta x)}{\Delta x} \rightarrow 0$
as $\Delta x \rightarrow 0$.

$\Rightarrow$ • if $\Delta x > 0$ and suff small, then
 $f(x) = f'(x_e) \cdot \Delta x + R(\Delta x) > 0$,

So $\Delta x(t) = x(t) - x_e$ is increasing in this case.

- And if $\Delta x < 0$ and suff small in magnitude, then $f(x) = f'(x_e) \cdot \Delta x + R(\Delta x) < 0$,

So $\Delta x(t)$ is decreasing in this case.

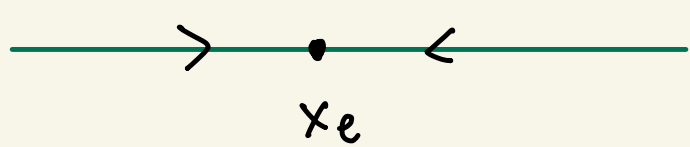


2. $f'(x_e) < 0$

$$f(x) - \overset{0}{f(x_e)} = f'(x_e) \Delta x + \overset{\text{h.o.t.}}{R(\Delta x)}$$

- If $\Delta x > 0$ & suff small, then $f(x) < 0$, so Δx is decreasing.

- If $\Delta x < 0$ & suff small in magnitude, then $f(x) > 0$, so Δx is increasing.


 $\Rightarrow x_e$ is asymptotically stable.

3. $f'(x_e) = 0$ and $f''(x_e) > 0$ h.o.t.

$$f(x) - \underbrace{f(x_e)}_0 = \frac{f''(x_e)}{2} (\Delta x)^2 + \underbrace{S(\Delta x)}_{\text{h.o.t.}}$$

• If $\Delta x \neq 0$, then $f(x) > 0$



where e.g.

$$\frac{S(\Delta x)}{(\Delta x)^2} \rightarrow 0 \quad \text{as } \Delta x \rightarrow 0$$

$\Rightarrow$ unstable ("semi-stable", but still unstable)

Linear ODEs

LTI

Def

A

linear (time-invariant)

ODE on $\mathbb{R}^n$ is:

$$\dot{x} = f(x)$$

where $x(t) \in \mathbb{R}^n$, and $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear map. Hence $f(x) = Ax$ for some $n \times n$ real matrix $A \in \mathbb{R}^{n \times n}$, so equivalently

$$\dot{x} = Ax.$$

Ex

Consider the 2nd-order linear ODE IVP on $\mathbb{R}$:

$$\ddot{z}(t) - 3\dot{z}(t) + 4z(t) = 0, \quad z(0) = z_0, \quad \dot{z}(0) = \dot{z}_0.$$

Let $x := \begin{bmatrix} z \\ \dot{z} \end{bmatrix} \in \mathbb{R}^2$. Then $\dot{x} = \begin{bmatrix} \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} \dot{z} \\ 3\dot{z} - 4z \end{bmatrix},$

or $\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ -4 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} z \\ \dot{z} \end{bmatrix}}_x$, so $\dot{x} = Ax$, and $x(0) = \begin{bmatrix} z_0 \\ \dot{z}_0 \end{bmatrix}$.

Moving toward the Fundamental Solution Thm for
LTI ODE IVP on $\mathbb{R}^n$ $\dot{x} = Ax$, $x(0) = x_0 \in \mathbb{R}^n$

• Special case: $n=1$. Then $A = a \in \mathbb{R}$,

and the linear ODE IVP

$$\dot{x} = ax \text{ w/ } x(0) = x_0,$$

w/ solution $x(t) = e^{at} x_0$

$$\frac{dx}{dt} = ax \Rightarrow \frac{dx}{x} = a dt$$

$$\ln x = at + C$$

$$x = \tilde{C} e^{at}$$

$$x(0) = \tilde{C} e^0 = x_0.$$

Note

$$e^{at} = 1 + at + \frac{(at)^2}{2!} + \frac{(at)^3}{3!} \dots$$
$$= \sum_{k=0}^{\infty} \frac{(at)^k}{k!}$$

Exponential of $A \in \mathbb{R}^{n \times n}$

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!}$$
$$= \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

Fact

the series $\sum_{k=0}^{\infty} \frac{A^k}{k!}$ converges,

and similarly $\sum_{k=0}^{\infty} \frac{(At)^k}{k!}$ converges $\forall t \in \mathbb{R}$.

Fact

for any $T > 0$, $\sum_{k=0}^{\infty} \frac{(At)^k}{k!}$ is uniformly convergent on $[-T, T]$.

Def

$$e^A := \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

Ex

Let $A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}$ be diagonal,

Then $At = \begin{bmatrix} a_{11}t & 0 \\ 0 & a_{22}t \end{bmatrix}$

$$A^2 t^2 = \begin{bmatrix} a_{11}^2 t^2 & 0 \\ 0 & a_{22}^2 t^2 \end{bmatrix},$$

$$A^3 t^3 = \begin{bmatrix} a_{11}^3 t^3 & 0 \\ 0 & a_{22}^3 t^3 \end{bmatrix}, \dots,$$

$$A^k t^k = \begin{bmatrix} a_{11}^k t^k & 0 \\ 0 & a_{22}^k t^k \end{bmatrix}.$$

$$\Rightarrow e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = \begin{bmatrix} \sum_{k=0}^{\infty} \frac{a_{11}^k t^k}{k!} & 0 \\ 0 & \sum_{k=0}^{\infty} \frac{a_{22}^k t^k}{k!} \end{bmatrix} = \begin{bmatrix} e^{a_{11}t} & 0 \\ 0 & e^{a_{22}t} \end{bmatrix}.$$

Ex

Consider a nilpotent matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Note: $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ z \\ 0 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Since $A^3 = 0$, $A^k = 0 \quad \forall k \geq 3$, so

$$e^{At} = I + At + \frac{(At)^2}{2} = \begin{bmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

More generally, for any nilpotent square matrix A w/ $A^k = 0$ for some $k \in \mathbb{N}$, $e^{At} = \sum_{j=0}^{k-1} \frac{(At)^j}{j!}$.

Properties of e^A and e^{At}

Note: if $A=0$, then $e^A = I$ and $e^{At} = I \forall t \in \mathbb{R}$.

Def Given $A, B \in \mathbb{R}^{n \times n}$, the commutator of A, B is $[A, B] = AB - BA$.

Note: A, B commute ($AB = BA$) $\Leftrightarrow [A, B] = 0$.

Fact Assume that $A, B \in \mathbb{R}^{n \times n}$ commute,
Then $e^{A+B} = e^A e^B$.

Corollary

for any $A \in \mathbb{R}^{n \times n}$ and $t, s \in \mathbb{R}$,
$$e^{At} e^{As} = e^{A(t+s)}.$$

pf

$$At \ As = ts \ A^2 = As \ At.$$

Fact

for any $A \in \mathbb{R}^{n \times n}$ and $t \in \mathbb{R}$, e^{At}
is invertible; in fact $(e^{At})^{-1} = e^{(-At)}.$

pf

$$I = e^{0 \cdot A} = e^{(t-t) \cdot A} = e^{tA} e^{-tA}.$$

Thm

Let $A \in \mathbb{R}^{n \times n}$. Define $F: \mathbb{R} \rightarrow \mathbb{R}^{n \times n}$
by $F(t) := e^{At}$. Then F is differentiable
everywhere, and

$$F'(t) = \frac{d}{dt} e^{At} = A e^{At} = e^{At} A \quad \forall t \in \mathbb{R}.$$

Pf

Fix any $t \in \mathbb{R}$. Consider

$$F(t+h) - F(t) = e^{A(t+h)} - e^{At}$$

$$= e^{Ah} e^{At} - e^{At}$$

$$= (e^{Ah} - I) e^{At}$$

Thus, $F'(t) = \lim_{h \rightarrow 0} \frac{F(t+h) - F(t)}{h}$

$$\lim_{h \rightarrow 0} \frac{e^{At} (e^{Ah} - I)}{h} \quad \Downarrow \quad \lim_{h \rightarrow 0} \frac{(e^{Ah} - I) e^{At}}{h}$$

$$e^{At} \left[\lim_{h \rightarrow 0} \frac{e^{Ah} - I}{h} \right]$$

$$\left[\lim_{h \rightarrow 0} \frac{e^{Ah} - I}{h} \right] e^{At},$$

so it suffices to show that $\lim_{h \rightarrow 0} \frac{e^{Ah} - I}{h} = A$.

Have

$$e^{Ah} - I = Ah + \sum_{k=2}^{\infty} \frac{(Ah)^k}{k!} = Ah + h^2 \sum_{k=2}^{\infty} \frac{A^k h^{k-2}}{k!}, \quad \text{so}$$

$$\frac{1}{h}(e^{Ah} - I) = A + h \sum_{k=2}^{\infty} \frac{A^k h^{k-2}}{k!}.$$

convergent and
uniformly bounded for,
say, $|h| \leq 1$.

Thus, $\lim_{h \rightarrow 0} \frac{1}{h}(e^{Ah} - I) = A$, as desired. $\square$

